

Final Project Roadmap — D&O Policy Analysis and Comparison Platform

Insurminds group · InsurMinds Course / I2A2 [Leia em português](ROTEIRO_PROJETO_FINAL.pt-BR.md)
Deadline: October 6, 2026 at 11:59 PM — this document was written on September 14, 2026; **22 days** remain.

What changes relative to Challenge 5

Three new things, and all three carry weight:

What's new	Why it needs attention
Pitch deck (InsurMinds_Projeto_Final.pptx)	Mandatory, literal filename. This is not a supporting slide: it is a graded deliverable
Video, 5 minutes maximum (InsurMinds_Projeto_Final.mp4)	Needs a working product to record. Leave it to the end and it won't exist
Unstructured input document	In Challenge 5 the API handed back ready-made numbers. Here the input is a legal PDF dozens of pages long

And a change in kind: **this is the course's final assessment**. Without it the group does not pass and nobody receives the advanced-module certificate.

Phase 0 — Decisions to make before writing code

None of this is technical enough for one person to decide alone. There are four decisions, and every one of them blocks work in more than one workstream.

F0.1 — Which fields to compare

The most important decision in the project. A D&O policy has dozens of clauses; the system doesn't need to extract all of them, it needs to extract the ones that change a decision.

Candidates for the discussion: maximum indemnity limit, deductible (retention), policy period, geographic scope, definition of the insured, main exclusions, defence-costs coverage, retroactive date, extended reporting period.

Who decides: Paulo Henrique leads — he's a broker, and this is exactly the reading he does at work. The group confirms the scope. **Target:** 8 to 12 fields. Fewer than that doesn't demonstrate comparison; more turns into shallow extraction across the board.

F0.2 — Where to get the policies

The assignment suggests public documents: templates published by insurers, SUSEP material, standard market clauses. **We need at least three real policies from three different insurers** — comparing two versions of the same template demonstrates nothing.

Who decides: Paulo Henrique points at where to look; anyone can download them. **Note:** the source of each document has to be cited in the report.

F0.3 — How to read the document

Three routes, and the choice reshapes all of workstream A:

1. **Native PDF** (embedded text) — `pypdf` handles it, it's fast and it costs nothing.
2. **Scanned PDF** — requires OCR (Tesseract locally, or a cloud service).
3. **Multimodal LLM** — send the page as an image and let the model read it.

Recommendation: try route 1 first and fall back to OCR only when a page has no text. Most public policy templates are native PDFs. But **the assignment requires accepting images**, so route 2 has to exist even if only for one document.

F0.4 — Where to store what was extracted

The assignment asks for "structured storage" and mentions SQL or NoSQL. For an MVP, SQLite handles it and requires standing up no service at all.

Recommendation: SQLite. The question to answer in the report isn't "which database", it's "why this database is enough for this problem".

The five workstreams

Workstream A — Ingestion and text extraction

Turn the incoming file into reliable text, knowing which page each passage came from — without that, workstream B's traceability dies.

Task	What it delivers
A.1	Document intake: accepts PDF and image, rejects everything else with a clear message
A.2	Native-PDF extraction, preserving the page number
A.3	OCR for pages with no embedded text
A.4	Fault tolerance: a corrupt document or an unreadable page doesn't take down the batch

Done when: the three policies chosen in F0.2 become text, and every passage can say which page it came from.

Workstream B — Clause extraction (the heart of the project)

This is where Generative AI genuinely comes in, and it's what the assignment grades as "correct use of Generative AI".

Task	What it delivers
B.1	Dictionary of the F0.1 fields: name, what it means, how it usually appears in the text
B.2	LLM extraction agent: takes the text, returns the structured fields
B.3	Traceability: every extracted field records its source page and passage
B.4	Validation: a missing field is an explicit <code>null</code> , never invented

Done when: running over the three policies produces a filled-in structure, and for any field it's possible to point at where it was in the document.

The lesson from Challenge 5, applied here: there, the guardrail stopped the model from quoting a number the forecast didn't contain. Here the stakes are higher — an invented indemnity limit is worse than a wrong message. **B.3 is not decoration: it's what separates this project from a well-formatted guess.**

Workstream C — Storage and comparison

Task	What it delivers
C.1	Database schema and persistence of processed policies
C.2	Comparison engine: field by field, across two or more policies
C.3	Difference classification: identical, different, missing in one of them
C.4	Comparative report in prose, generated by an LLM from the differences

Done when: given two policies, out comes a table of differences plus a summary explaining what they mean for the buyer.

Mind the boundary: C.2 decides *what* is different (deterministic, testable); C.4 explains *why it matters* (LLM). Mixing the two was the mistake that cost us a consolidation PR in Challenge 5.

Workstream D — Interface and demo

Task	What it delivers
D.1	Policy upload through the interface
D.2	Side-by-side view of the differences
D.3	Traceability on screen: click a field and see where it came from
D.4	Command-line demo, so the video doesn't depend on the interface

Done when: you can upload two policies and see the comparison without touching a terminal.

Workstream E — Documentation, pitch and video

Bigger than in Challenge 5, and with two deliverables that aren't text.

Task	What it delivers
E.1	README with the six items the assignment requires
E.2	Technical report with seven sections (two new: known limitations and future work)
E.3	<code>InsurMinds_Projeto_Final.pptx</code> — pitch deck
E.4	<code>InsurMinds_Projeto_Final.mp4</code> — video, 5 minutes maximum
E.5	Code ZIP and organisation of the <code>Projeto_Final_Artefatos</code> folder

Suggested split

Based on what each person demonstrated in Challenge 5. **Juliana is travelling**, so workstream E has been divided among the four of us — she stays on as the group representative, responsible for submission, and picks back up whatever makes sense when she returns.

Member	Workstream	Why
Paulo Henrique	F0.1, F0.2, B.1 + the business content of the pitch	He's the broker. The D&O field dictionary is this project's <code>regras.yaml1</code> — the piece that gave Challenge 5 its strongest argument. And the pitch's problem statement is the pain he lives: specialist hours spent comparing clauses
Nicole Paes	Workstream B (B.2 to B.4)	She delivered the agents and found the LLM bug that was disguising itself as a correct fallback
Daniel Ramon	Workstreams A and C + E.1, E.5 + consolidating the report	Collection and the rules engine were his in Challenge 5, and so were the README and the packaging
Paulo Roberto	Workstream D + E.3 (pitch) + E.4 (video)	He built the interface and the demo; he's the one who best knows how to show the system working
Juliana Catarina	Submitting the delivery	Group representative — the submission must go out from her email

The technical report without Juliana

E.2 is the largest orphaned piece. Instead of a single owner, **each workstream writes the section that corresponds to it** and Daniel consolidates:

Report section	Who writes it
Solution architecture	Daniel
Tech stack	Daniel
Description of the agents	Nicole (workstream B) and Paulo Roberto (workstream D)
Processing flow	Daniel
Rationale for the architectural decisions	whoever made each decision, in one sentence
Known limitations	everyone — each of us knows where our own workstream is fragile
Future work	everyone

When Juliana returns, the natural move is to hand her back the review and final formatting of the report — that's what she did well in Challenge 5, and it's end-of-cycle work. If she's back in time, the pitch can go to her as well, freeing Paulo Roberto to focus on the video. Reassess around **September 27**.

Why the video goes to Paulo Roberto: recording requires the application running in the hands of whoever built it. He assembles the pitch and records; Paulo Henrique supplies the business-problem content, which is the first part of the deck.

Suggested calendar

By	What has to be ready
Sep 17	Phase 0 closed. Policies downloaded, fields defined, stack chosen
Sep 22	Workstreams A and B working: a document goes in, structured fields come out
Sep 27	Workstream C: comparison between two policies producing output. Reassess what to hand back to Juliana
Sep 30	Workstream D: demonstrable interface. Feature freeze
Oct 2	Video recorded and pitch ready
Oct 4	Final report and ZIP. Group review
Oct 6	Delivery — do not leave it for this day

The September 30 freeze isn't bureaucracy: **without a stable product there is no video**, and the video is the deliverable easiest to lose for lack of time.

Known risks

1. **The video slipping to the end.** It's the item most likely to be late. That's why it has its own owner and its own date.
1. **Hallucinated extraction.** An invented indemnity limit slides past unnoticed in a demo and destroys credibility the moment someone asks about it. B.3 and B.4 exist against exactly this.
1. **LLM quota.** It already bit us in Challenges 4 and 5. Processing a whole policy burns far more tokens than generating an SMS — **test consumption early**.
1. **Difficult policies.** If all three documents turn out to be scanned and illegible, workstream A becomes the bottleneck. That's why F0.2 comes before everything else.
1. **Four people instead of five.** With Juliana travelling, workstream E is split and the report has no single owner. The risk isn't the writing — it's nobody noticing that a section ended up without an author. The section table above exists for that; check it at the October 4 review, and **don't count on her return** for the delivery to happen.